Team 354
Problem A: Ion Thrusters to Saturn

November 6, 2020

Abstract

This paper investigates the optimization of the trajectory of spacecraft equipped with low-thrust ion engine which provides the constant thrust F_T of 400 mN, with the specific impulse I_{sp} of 4000 s. The main goal is to efficiently send the spacecraft from the circular orbit of radius 6650.32 km (equivalent to the time period of 90 minutes) around the earth to the circular orbit of radius 271033.49 km (equivalent to the time period of 40 hours) around the Saturn. This was done using the combined method of ion propulsion and Gravity Assist (GA). We constructed the model which simulates the trajectory of the spacecraft based on Newton's and Kepler's law of gravitation. With this model, the spacecraft took 9 years and 73 days to travel from the initial configuration to Saturn. The optimal amount of fuel consumed in this model was found to 1808.75 kg considering the total initial mass of the spacecraft (including the fuel) to be 5000 kg. The method of controlling the thruster during the journey is discussed below.

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1 Notations Used

Table 1: Table of notation

Symbols	Meaning	Numerical Value
G	Universal Gravitational Constant	6.67408×10^{-11} m^3 kg^{-1}s^{-2}
M_E	Mass of Earth	5.972×10^{24} kg
M_S	Mass of Saturn	5.683×10^{26} kg
m_s	Instantaneous mass of spacecraft	-
a	Semi-major axis length	-
T	Time period of revolution	-
$r_{s,E}$	Radius of initial orbit of spacecraft	6650.32 km (calculated)
$r_{s,S}$	Radius of final orbit of spacecraft	271033.46 km (calculated)
$v_{s,E}$	Velocity in the initial orbit of spacecraft	7.738 km s^{-1} (calculated)
$v_{s,S}$	Velocity in the final orbit of spacecraft	11.826 km s^{-1} (calculated)
F_T	Constant thrust on the spacecraft	400 mN (given)
g_0	Acceleration due to gravity on Earth	9.81 ms^{-2}
I_{sp}	Specific impulse	4000 s (given)
δ	Turning angle	-

2 Introduction

Ion thruster (a.k.a Kaufman Ion Engines) is electric propulsion system used for the spacecraft in several inter-planetary and deep space missions. The underlying principle behind it is the production of thrust by accelerating the ions using electric field via exhaust. Because of its mass and inert behaviour, Xenon (Xe) is used as the common propellant for the ion propulsion system. Recently ion thrusters have been designed for various missions - from keeping the satellites orbiting the earth in their proper position to propelling the spacecraft to different planets (for example: Mars, Jupiter, Saturn) in the solar system.

The concept of electrostatic thrusters dates back to late 1950's with the first demonstration on July 20, 1964 on a sub-orbital spaceflight of the Space Electric Rocket Test 1 (SERT I) spacecraft. In 2007, NASA launched the DAWN spacecraft which travelled deep into the solar system utilizing the three NASA Solar Technology Application Readiness (NSTAR) ion thrusters [NASb].

But why are ion thrusters becoming more popular recently? Well, the answer lies behind various factors. Ion thursters require less propellant than its traditional counterpart chemical propellant rocket engines. They have comparatively higher specific impulse - the quantity that shows how efficiently the propellant is used - which is most suitable for long-duration missions. Although the thrust produced is low, they can achieve incredibly high speed gradually.

Due to the high efficiency and applicability towards deep space long-duration travel, researchers are looking for various ways to optimize the trajectory, cost, fuel, and the time of flight. One of the best optimization technique would be using the Gravity Assist (GA) along

with ion propulsion which is described in this paper. Also, the way the thruster is controlled must be taken into account for the optimization.

3 Theory

3.1 Law of Gravitation and Kepler's Laws

In the scenario of the problem, the initial and final states of the spacecraft are circular orbits around the Earth and Saturn, respectively. Solving for the period and speed of the spacecraft in this orbit requires knowledge about the forces acting on the spacecraft, primarily the force of gravity. When a spacecraft of mass m_s is orbiting around a body of much larger mass M, the force of gravity between the body and the spacecraft is given by Newton's Universal Law of Gravitation:

$$\vec{F} = G\frac{Mm_s}{r^2}\hat{r} \tag{1}$$

where r is the distance between the large body and the spacecraft, and G is the universal gravitational constant. The unit vector $\hat{r}$ points from the center of mass of M to the spacecraft. By Newton's Second Law, the gravitational force acting on the object of mass m_s orbiting a body of mass $M \gg m_s$ is equal to its centripetal force:

$$G\frac{M_s m}{r^2} = \frac{m_s v^2}{r} \tag{2}$$

where v is the velocity of the spacecraft in circular orbit. We can rearrange this equation to obtain an expression for the orbital velocity in terms of the the distance of the spacecraft from the center of mass of the large body:

$$v = \sqrt{\frac{GM}{r}} \tag{3}$$

The motion of the planets around the solar system is described by Kepler's Laws, which state that each planet moves around the sun in an elliptical orbit. In particular, Kepler's Third Law relates the semi-major axis of the planet's orbit to the period of its revolution around the sun:

$$T^2 = \frac{4\pi^2}{GM}a^3 \tag{4}$$

Here, a is the semi-major axis of the planet's orbit, T is the period of revolution, G is the universal gravitational constant (approx 6.674 $\mathrm{m^3/kg{\cdot}s^2}$). Rearranging, an expression for a depending on T can be obtained, with all other quantities being constant:

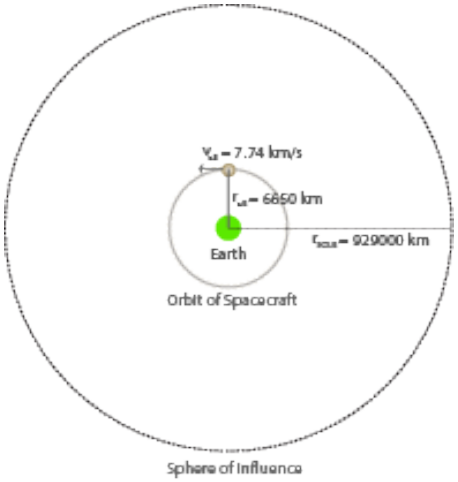
$$a = \left(\frac{T^2 GM}{4\pi^2}\right)^{\frac{1}{3}}$$

When the orbit is circular, the radius r of the orbit takes the place of the semi-major axis a. For a circular orbit around the Earth with a period of 90 minutes:

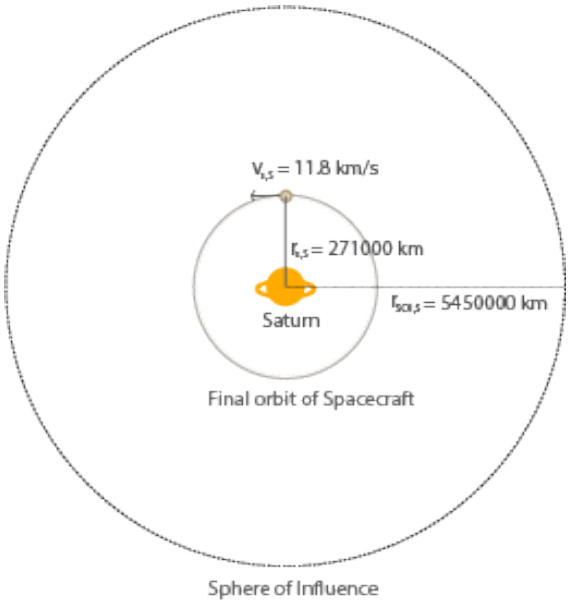
$$r_{s,E} = 6650.321 \text{ km} \qquad v_{s,E} = 7.738 \text{ km/s}$$

For a circular orbit around Saturn with a period of 40 hours:

$$r_{s,S} = 271033.462 \text{ km} \qquad v_{s,S} = 11.826 \text{ km/s}$$



(a) Initial Orbit of the Spacecraft around Earth



(b) Final Orbit of the Spacecraft around Saturn

Figure 1: Orbit of Spacecraft around Earth and Saturn. [Figures not to be scaled]

3.2 Specific Impulse

The specific impulse is a measure of how much thrust a spacecraft can achieve given a particular rate of fuel consumption. These quantities are related to each other by the following equation:

$$F_T = g_0 \cdot I_{sp} \cdot \dot{m} \tag{5}$$

where I_{sp} is the specific impulse, F_T is the thrust force on the spacecraft, g_0 is the acceleration due to gravity on the Earth's surface (≈ 9.81 m/s^2), and $\dot{m}$ is the instantaneous rate of change in the mass of the spacecraft.

3.3 Orbital spiral maneuver (SM) with low thrust

For transferring between orbits around the central body, the low thrust engines may be employed. The shape of the trajectory will be "spiral", hence the name, and independent of the steering angle. The simplest model carried out in this paper keeps the thrust applied normal to radius vector, giving the upper bound for fuel consumption during this maneuver. Some optimization of thrust angle can be done such that the change of distance from central object is maximized, and fuel efficiency increased.

The spiral trajectory is calculated based on the following equations. Consider the net force experienced by the spacecraft when the ion thruster is turned on. It is given by:

$$\vec{F}_{net} = -G\frac{M_E m_s}{r^2}\hat{e}_r + F_T\hat{e}_\phi \tag{6}$$

In terms of Cartesian coordinate system, the polar unit vectors can be represented as:

$$\begin{aligned} \hat{e}_r &= \cos(\phi)\hat{x} + \sin(\phi)\hat{y} \\ \hat{e}_\phi &= -\sin(\phi)\hat{x} + \cos(\phi)\hat{y} \end{aligned} \tag{7}$$

where ϕ is the angle made by radial vector with the x-axis.

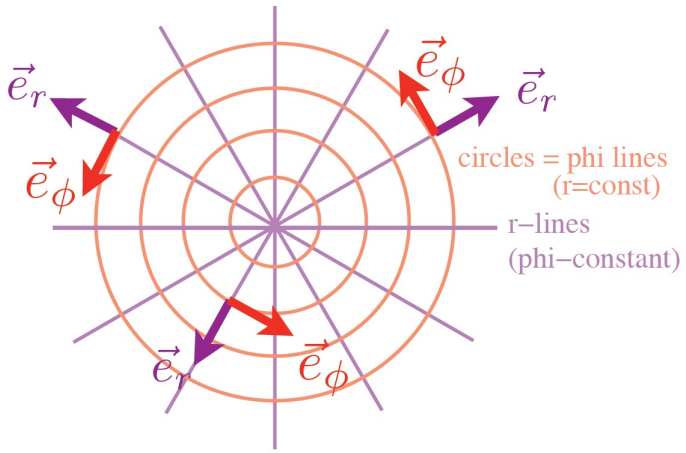


Figure 2: Polar coordinate system [Phy]

Similarly, $\vec{F}_{net}$ is given by:

$$\vec{F}_{net} = \frac{dm_s}{dt}\frac{dr(t)}{dt} + m_s\frac{d^2r(t)}{dt^2} \tag{8}$$

where m_s also changes with time due to fuel exhaust.
Hence, combining equation 6 and 8, we get two differential equations:

$$\begin{pmatrix} \dot{m}_s\dot{x}(t) + m_s\ddot{x}(t) \\ \dot{m}_s\dot{y}(t) + m_s\ddot{y}(t) \end{pmatrix} = \begin{pmatrix} -G\frac{M_E m_s}{r^2}\cos\phi - F_T\sin(\phi) \\ -G\frac{M_E m_s}{r^2}\sin\phi + F_T\cos(\phi) \end{pmatrix} \tag{9}$$

where $\dot{}$ represents the first time derivatives, and $\ddot{}$ represents the second time derivatives.
Using equation 5, the value of $\dot{m}_s = -\dot{m}$ is calculated to be $\approx 1 \times 10^{-5}$ kg/s. Neglecting the effect of $\dot{m}_s$ in equation 9, we can write:

$$\begin{pmatrix} m_s\ddot{x}(t) \\ m_s\ddot{y}(t) \end{pmatrix} = \begin{pmatrix} -G\frac{M_E m_s}{r^2}\cos\phi - F_T\sin(\phi) \\ -G\frac{M_E m_s}{r^2}\sin\phi + F_T\cos(\phi) \end{pmatrix} \tag{10}$$

Hence, the final equation of motion for the spacecraft exhibiting spiral maneuver (SM) is given by:

$$\begin{pmatrix} \ddot{x}(t) \\ \ddot{y}(t) \end{pmatrix} = \begin{pmatrix} -G\frac{M_E}{r^2}\cos\phi - \frac{F_T}{m_s(t)}\sin(\phi) \\ -G\frac{M_E}{r^2}\sin\phi + \frac{F_T}{m_s(t)}\cos(\phi) \end{pmatrix} \tag{11}$$

3.4 Gravity Assist (GA) maneuvers

Gravitational slingshot is method used to alter path and direction of a spacecraft, by taking advantage of relative moment and gravity of planet or Sun. The idealized equation which gives us the gain in velocity is described below:

$$v_f = (v_i + 2u) \cdot \sqrt{1 - \frac{4uv_i(1 - \cos\theta)}{(v_i + 2u)^2}} \tag{12}$$

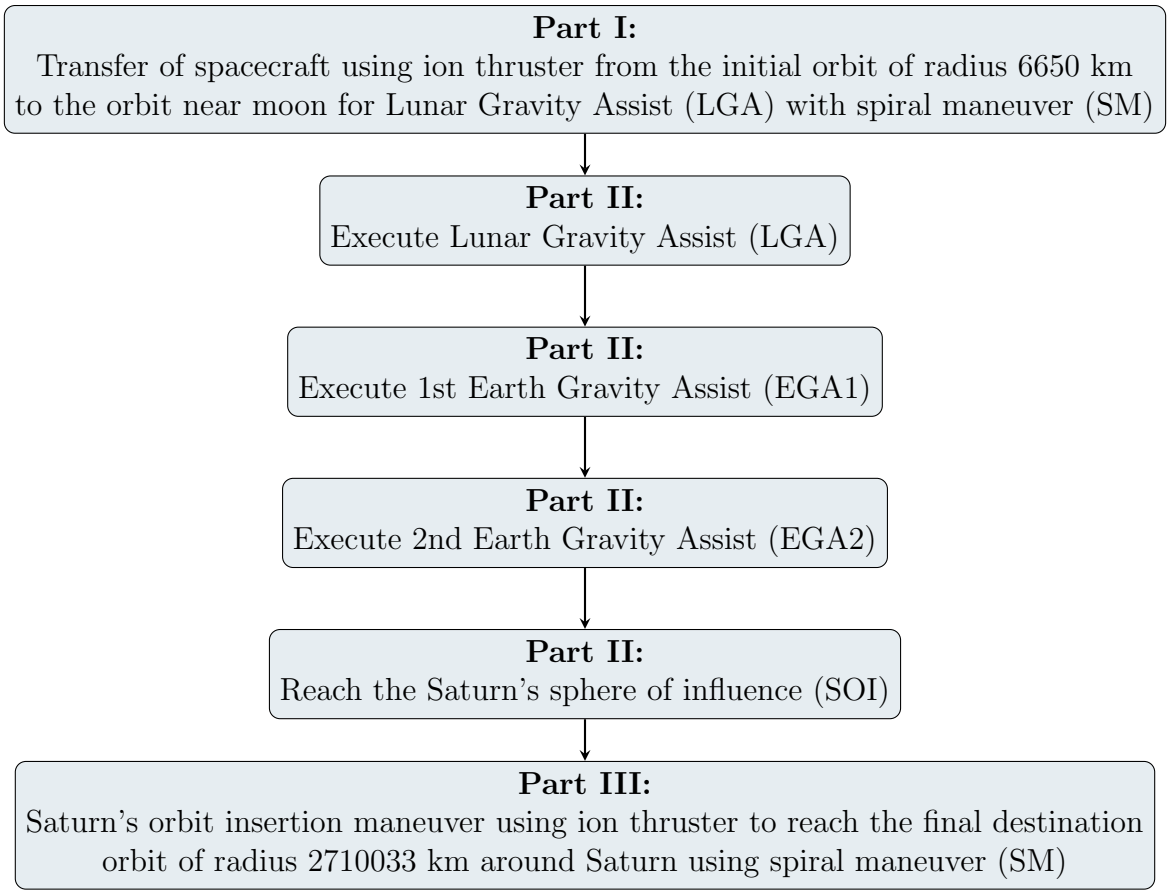
Initial and final velocity of spacecraft are represented by v_i and v_f respectively, u is relative speed of larger body with respect to spacecraft, and θ is approaching angle of spacecraft with respect to velocity of larger planet.
In the model above, the larger body was assumed to be a point, which may effect the θ after the slingshot effect. Another quantity, named turning angle is applied to consider all effects of the gravity assist:

$$\delta = 2\sin^{-1}\left(\frac{1}{1 + \frac{r_p v_i}{GM}}\right) \tag{13}$$

In which r_p is distance between spacecraft and larger body, M mass of larger body and G gravitational constant.
Usage of initial angle of $\theta = 0$, in the idealized equation θ can be replaced by turning angle δ. This principle has been used for all gravity assist calculations.

4 Method

In order to send a spacecraft of mass 5000 kg employing ion thrust propulsion (constant thrust = 400 mN, specific impulse = 4000 s) from a circular orbit on Earth of radius approx. 6650 km (equivalent to the time period of 90 minutes) to a circular orbit on Saturn of radius approx 271033 km, the joint method of ion thrust and gravity assists (GA) was applied. Therefore, the whole process is divided into three main parts. It is summarized in the following flowchart.



The Euler-Richardson numerical integration scheme was used to determine trajectories for the spiral maneuvers in parts I and III, as well as the Earth-to-Saturn interplanetary maneuver in Part II (SOI). Annotated code has been included in the Appendix. Similarly, python scripts were used to find the trajectory of spacecraft in part II which comprises gravity assists (GA) from moon and earth.

5 Results

5.1 Earth Spiral Maneuver

The trajectory of the spacecraft resulting from numerical integration is shown in Figure 3 below. Throughout the entire maneuver, the thrust force was applied in a direction perpendicular to the radius vector, going with the motion of the spacecraft.

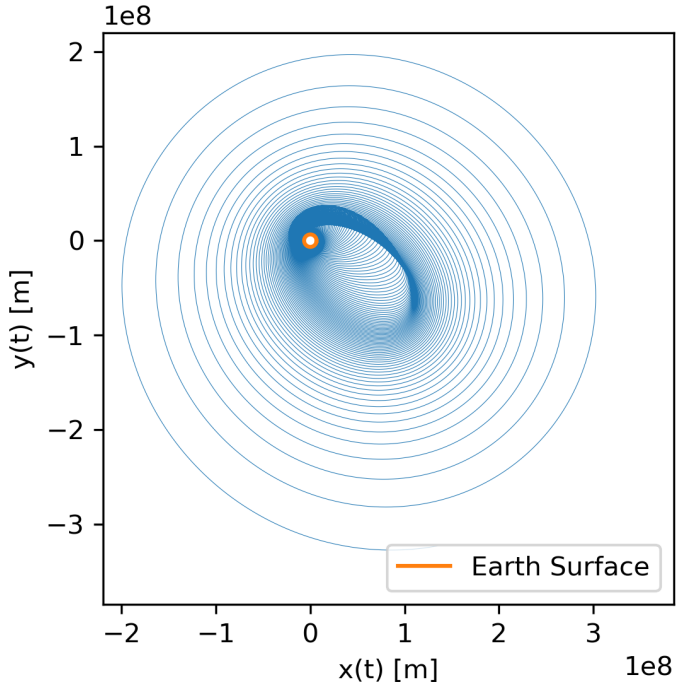


Figure 3: Spiral maneuver used to leave the Earth's sphere of influence. The orange line denotes the surface of the Earth. The total time of flight for the whole maneuver is 24054600.0 s (≈ 278 days).

Aside from the final speed, the speed of the spacecraft across the duration of the flight was also determined. The results were plotted and shown in Figure 4 below.

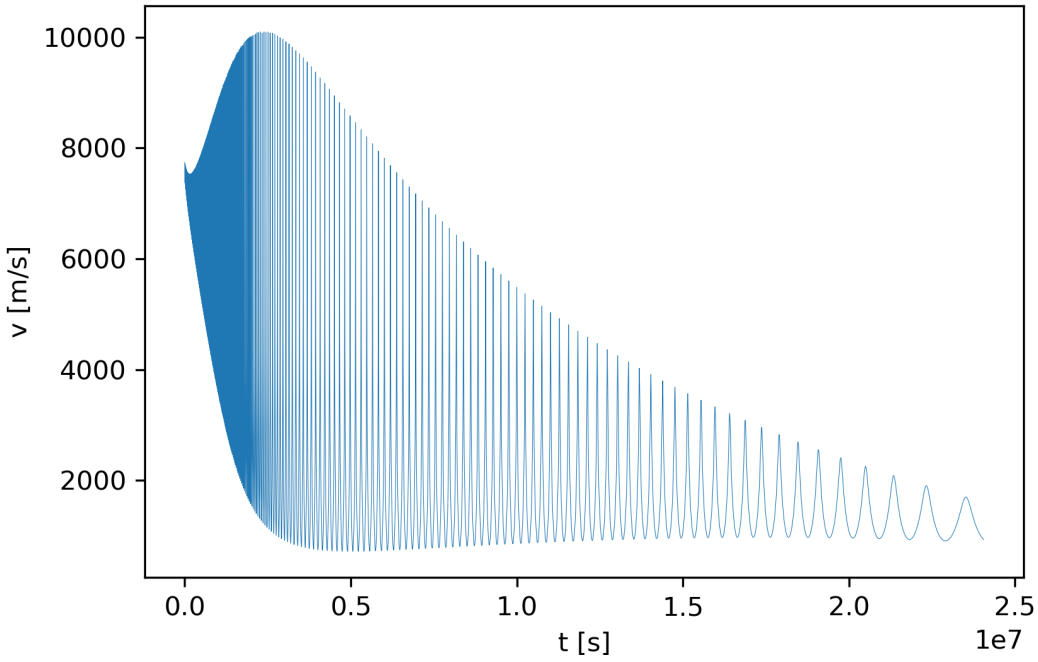


Figure 4: Speed vs. time of the spacecraft during the spiral maneuver.

5.2 Lunar and double Earth Gravity Assists

After reaching Moon's sphere of influence, the gravity assist could be applied. At distance 30000 km from center of Moon, with initial velocity 0.934 km/h, using equation 12, the velocity of spacecraft has been of 1.115 km/h has been reached.

With velocity reached from LGA, the spacecraft barely escapes the gravitational field of Earth, and has heliocentric orbit, with approximately Earth-Sun distance. One more engine burn has been used to face the spacecraft back to trajectory to Earth, and increase initial velocity for first Earth Gravity Assist. The engine was burned for $\approx 1/15$ of period.

First EGA was done such that the velocity of spacecraft of 5.4 km/h was obtained, which pushes it to heliocentric orbit several Sun-Earth distances. The engine burn has been applied, in order to adjust the position and initial velocity for second EGA.

After second EGA, the spacecraft has enough energy for inter planatary Saturn mission, with velocity with respect to Earth being 10.65 km/s.
The average time needed for each step is roughly estimated by Kepler's third law 4, and estimated timing was shown in Table 2.

5.3 Earth to Saturn Trajectory

The trajectory of the spacecraft just as it leaves the Earth's sphere of influence until it reaches Saturn's sphere of influence was solved for using numerical integration. The maneuver is shown in the Figure below. The reference frame used in this simulation is one where the sun is stationary, and the additional velocity that comes with moving from the stationary Earth frame to this new frame has been taken into account in the numerical integration.

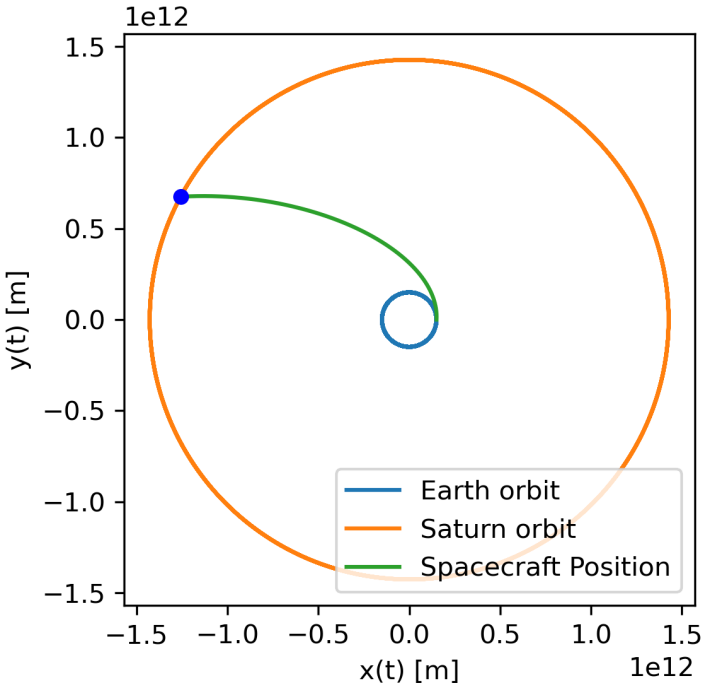


Figure 5: Speed vs. time of the spacecraft during the interplanetary maneuver. The total time of flight for this maneuver is approximately 3.5 years.

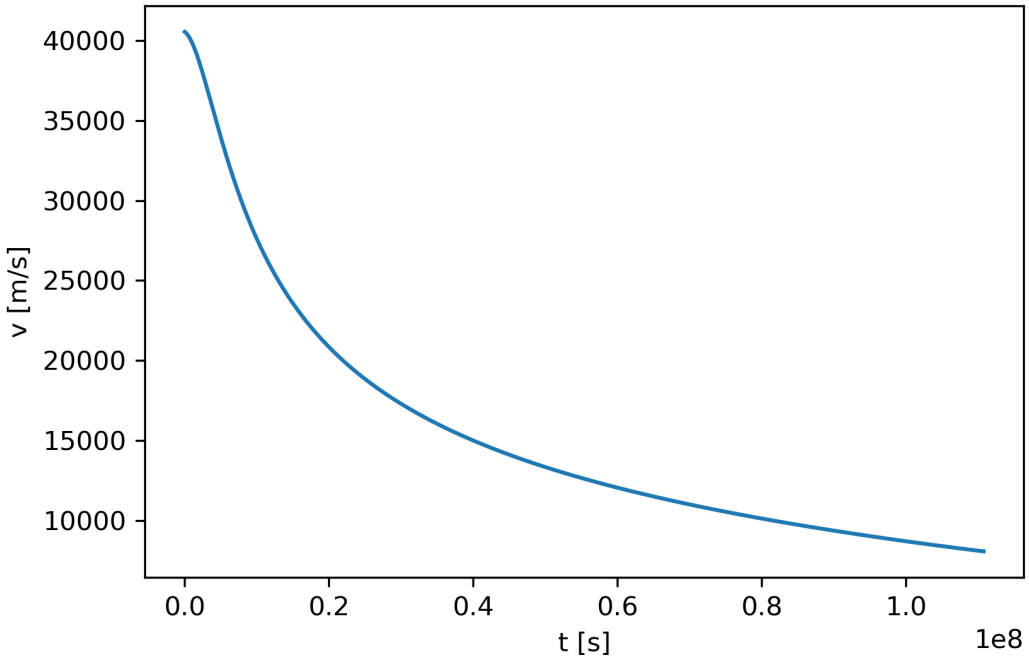


Figure 6: Speed vs. time of the spacecraft during the interplanetary maneuver.

5.4 Saturn Spiral Maneuver

The last numerical integration was conducted to simulate the insertion maneuver of the spacecraft into the desired circular orbit around Saturn. The results are given in figure 7. For this insertion trajectory, the thrust force was applied in a direction perpendicular to the radius vector, against the motion of the spacecraft.

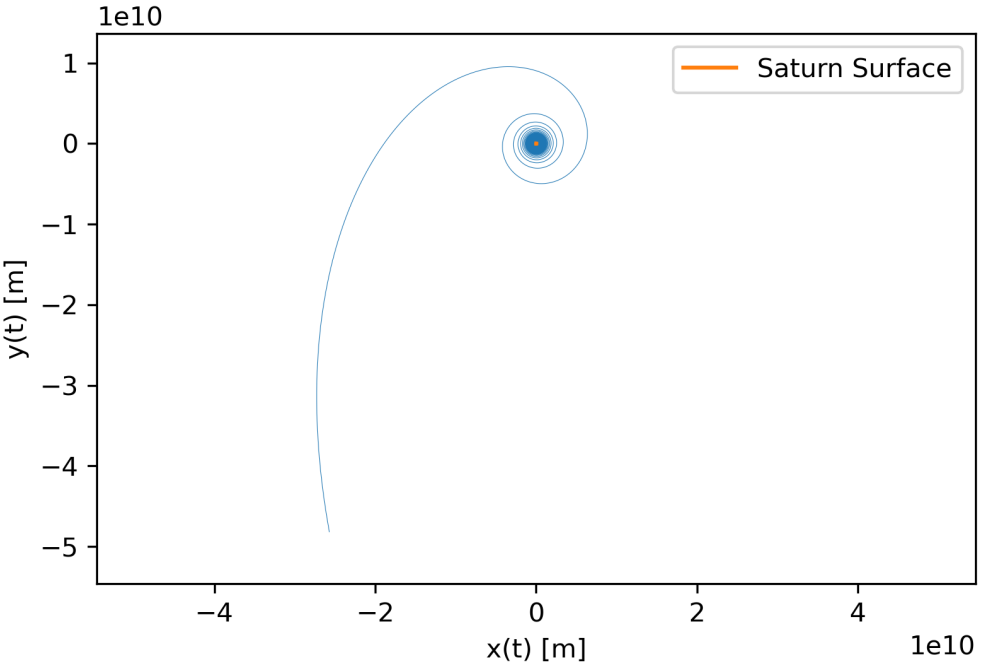


Figure 7: Speed vs. time of the spacecraft during the interplanetary maneuver.

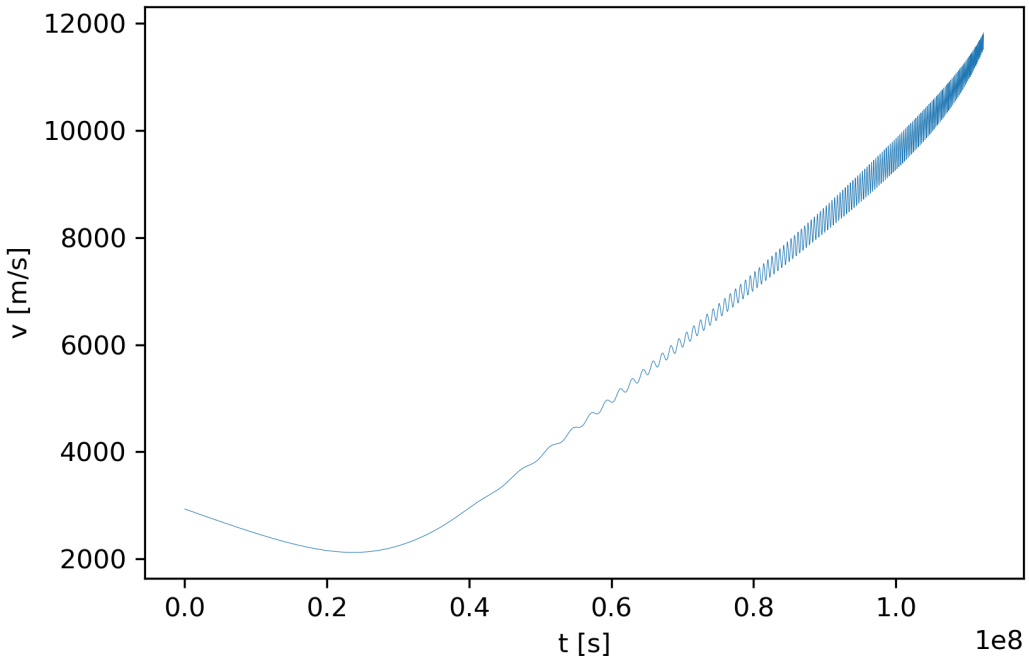


Figure 9: Speed vs. time of the spacecraft during the interplanetary maneuver.

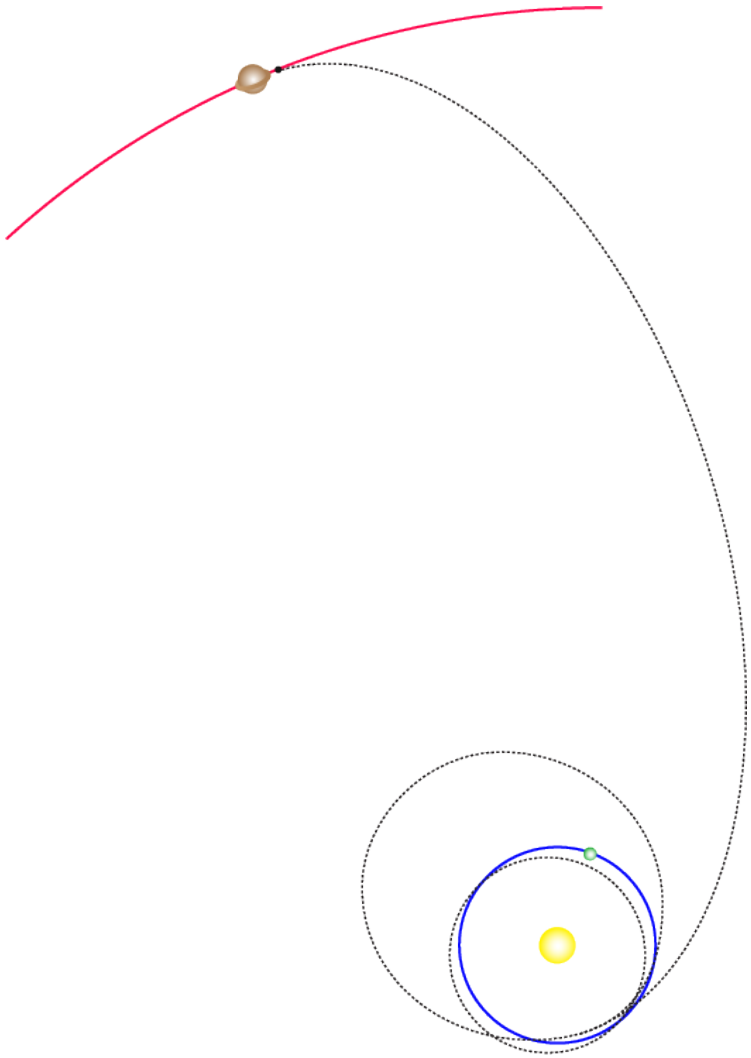


Figure 8: Multi-gravity assist transfer from Earth to Saturn using one LGA and two EGAs. Lunar gravity assist is relatively small and not shown in the picture. The blue orbit represents Earth's orbit around sun (yellow dot in the middle) and the red orbit represents the orbit of Saturn. The black dotted trajectory is the path followed by spacecraft. (*The figure is not be scaled.)

From the above calculations and numerical integration, the total change in mass of the spacecraft at each phase of the mission, along with the cumulative time duration to complete each phase and the velocity of the spacecraft at that point are given in Table 2.

Table 2: Phase of the mission with respect to estimated time since launch, with mass of spacecraft and velocity of spacecraft at beginning of each phase.

Phase	Estimated time	Mass of spacecraft [kg]	Velocity of spacecraft [km/s]
Earth transfer orbit	0	5000	7.738
Lunar approach	278 days	4759.45	0.934
After LGA	279 days	4759.45	1.115
Before first EGA	1 year, 60 days	4548.42	1.115
After first EGA	1 year, 60 days	4548.42	5.42
Before second EGA	2 years, 25 days	4451.95	5.45
After second EGA	2 years, 25 days	4451.95	10.65
Saturn approach	5 years, 208 days	4451.95	8.92
Circular Saturn orbit	9 years, 73 days	3191.25	11.83

6 Discussion and Conclusion

This mission objective was to put the 5000 kg spacecraft from 90 minute circular Earth orbit, to 40 hours circular orbit around Saturn, such that final mass is maximized, or in other words, the fuel usage was minimized. The spacecraft reached the circular Saturn orbit after 9 years and 73 days, and with the final mass being 3191.25 kg.

First method of Earth Spiral Maneuver with Low thrust engine was used in order to transfer from low-Earth circular orbit to a highly elliptical one. The sphere of influence of the Moon has been reached, and by using LGA maneuver, the spacecraft has barely escaped Earth's orbit. After some adjustments and occasional engine burns, two EGA maneuvers were used in order to get suitable velocity to reach the Saturn. With Saturn reach, the Saturn Spiral Maneuver was used, in order to put the spacecraft in circular orbit required.

By comparing Figures 3 and 7, it can be seen that the spiral trajectory of Saturn is much more spaced out than that of the Earth. This is to be expected. Since the distance that the spacecraft covers in one revolution is much larger for the case of a Saturn orbit, the thrust force has more time to change the total energy of the orbit. Figures 4 and 9 show that the velocity of the spacecraft is rapidly oscillating. This is due to the highly eccentric orbits caused by a thrust maneuver where the direction of the thrust is always perpendicular to the radius vector. In the case of the Earth-to- Saturn maneuver, the decrease in speed is much more uniform.

6.1 Limitations and Improvements

In this paper, the use of gravity assists was done to minimize the amount of fuel expelled by the spacecraft. This is, however, just one of many ways to optimize fuel expenditure. Another way to further optimize the trajectory is with the use of direct methods– numerical methods which obtain an optimal trajectory through continuously improving approximations. As an example, one popular method of trajectory optimization for low-thrust spacecraft is the method developed by Sims and Flanagan [Sim12]. Their method involves dividing a proposed trajectory into legs, separated by control nodes which are arbitrary destinations in the trajectory. By propagating forwards and backwards in time from each control node, an optimal trajectory can be reached. This is just one example of several "multiple shooting" direct methods for optimization. One more possible improvement is to simulate the entire trajectory of the spacecraft (i.e. from the circular orbit of the Earth to the circular orbit of Saturn). This would require precise knowledge about the configurations of the Earth, Moon, and Saturn for the entire duration of the mission. A better trajectory computation should be done which takes into account the precise angle at which the spacecraft enters Saturn's sphere of influence (with respect to, for example, the direction of Saturn's motion), and the position of the Earth and angle at which the spacecraft leaves the Earth's sphere of influence.

To extend this research, it is possible to come up with feasible launch dates for this proposed mission, which involves one Lunar gravity assist, and two Earth gravity assists to reach Saturn. Since the Saturn-Earth synodic period is small (1.1 years), there are potentially many possible opportunities to execute this mission.

To make the simulation more precise, the gravitational forces exerted by the other planets and Saturn's moons and its rings.

The most fuel used was in Spiral approach of Saturn. This fuel may be optimized better if Gravity Assist maneuver was used on one of the Saturn's moons.

For approach used in this paper, the thrust of ion engine was always perpendicular to radius vector of the spacecraft. However, may not be the most efficient way, and several steering laws can be applied such that fuel usage and time is minimized.

7 Appendix

```
import os
import math
import numpy as np
import matplotlib.pyplot as plt

sun = ephem.Sun()
saturn = ephem.Saturn()

#Spiral Maneuver (for leaving the Earth's Orbit)
#Purely angular thrust is applied.

#Initial Parameters
F_thrust = 0.4 #Newtons
I_s = 4000 #seconds (specific impulse)
m = 5000 #kg
T_i = 90*60 #seconds (inintial orbital period of spacecraft around Earth)
T_f = 40*3600 #seconds (final orbital period of spacecraft around Saturn)

#Solar System Constants
m_E = 5.972*10**24 #kg (mass of earth)
m_Sat = 5.683*10**26 #kg (mass of saturn)
m_Sun = 1.989*10**30 #kg (mass of sun)
R_E = 6.59E6 #m (radius of Earth)
G = 6.67*10**-11 #Nm^2/kg^2
g = 9.8 #m/s^2

T_Sat = 10759.22*86400 #seconds (orbital period of Saturn around Sun)
T_E = 365.25*86400 #seconds (orbital period of Earth around Sun)
#orbital radius of Saturn around Sun
R_Sat_orbit = (((T_Sat**2)*G*m_Sun)/(4*np.pi**2))**(1/3)
#orbital radius of Earth around Sun
R_E_orbit = (((T_E**2)*G*m_Sun)/(4*np.pi**2))**(1/3)

t_meas = 5 #time difference to be measured (in years)
t_step = 24*3600 #time step for the discretization (in seconds)
t_array = np.arange(0,t_meas*365*86400,t_step)
Sat_x = R_Sat_orbit*np.cos((2*np.pi/T_Sat)*t_array+theta_i_Sat)
Sat_y = R_Sat_orbit*np.sin((2*np.pi/T_Sat)*t_array+theta_i_Sat)
E_x = R_E_orbit*np.cos((2*np.pi/T_E)*t_array)
E_y = R_E_orbit*np.sin((2*np.pi/T_E)*t_array)


def Earth_Orbit():
```

```
    t_array = np.arange(0,1,0.01)
    x_E_surf = R_E*np.cos((2*np.pi*t_array))
    y_E_surf = R_E*np.sin((2*np.pi*t_array))
    return x_E_surf,y_E_surf


#Euler-Richardson Scheme to obtain trajectory of spacecraft
def ER_SC_Spiral_Earth(initial,F,t,t_stop):
    #initial parameters
    x,y,vx,vy,ax,ay=initial
    print(initial)

    #arrays of each variable
    x_array = np.array([])
    y_array = np.array([])
    vx_array = np.array([])
    vy_array = np.array([])
    ax_array = np.array([])
    ay_array = np.array([])
    t_array = np.array([])

    i=0
    m=5000
    m_r = 1e-5

    while (x**2+y**2)**0.5<=350E6:
        #compute acceleration
        ax = -G*(m_E*x)/(x**2+y**2)**1.5- F*(y/(x**2+y**2)**0.5)/m
        ay = -G*(m_E*y)/(x**2+y**2)**1.5+ F*(x/(x**2+y**2)**0.5)/m
        #calculate middle terms
        x_mid = x+t/2*vx
        y_mid = y+t/2*vy
        vx_mid = vx+t/2*ax
        vy_mid = vy+t/2*ay
        ax_mid = -G*(m_E*x_mid)/(x_mid**2+y_mid**2)**1.5 -
        F*(y_mid/(x_mid**2+y_mid**2)**0.5)/m
        ay_mid = -G*(m_E*y_mid)/(x_mid**2+y_mid**2)**1.5 +
        F*(x_mid/(x_mid**2+y_mid**2)**0.5)/m


        #append to arrays
        t_array = np.append(t_array,t*i)
        x_array = np.append(x_array,x)
        y_array = np.append(y_array,y)
        vx_array = np.append(vx_array,vx)
```

```
        vy_array = np.append(vy_array,vy)
        ax_array = np.append(ax_array,ax)
        ay_array = np.append(ay_array,ay)

        #update vx, vy, x, and y
        x = x + t*vx_mid
        y = y + t*vy_mid
        vx = vx + t*ax_mid
        vy = vy + t*ay_mid
        m -= m_r*t
        i=i+1

        print("radius:",(x**2+y**2)**0.5)


    print("Time of Flight: ", t_array[-1])
    print("Final Speed: ", (vx_array[-1]**2+vy_array[-1]**2)**0.5)
    print("Mass consumed:", 5000-m)
    return(t_array, x_array, y_array, vx_array, vy_array, ax_array, ay_array)

#position, velocity, acceleration values at circular orbit
x0_sc = 6650321.625#m (radius of orbit)
y0_sc = 0
vx0_sc = 0
vy0_sc = 7.738*1000#m/s (velocity determined using Kepler's Laws)
ax0_sc = 0
ay0_sc = 0
ini1 = [x0_sc,y0_sc,vx0_sc,vy0_sc,ax0_sc,ay0_sc]

arrays1 = ER_SC_Spiral_Earth(ini1,F_thrust,200,86400*1)

#Plotting
fig2 = plt.figure()
plt.plot(arrays1[1],arrays1[2],linewidth="0.25")
plt.plot(Earth_Orbit()[0],Earth_Orbit()[1],label = "Earth Surface")
plt.legend(loc ="lower right")
plt.xlim(-2.2E8, 1.1*350E6)
plt.ylim(-1.1*350E6, 2.2E8)
plt.xlabel("x(t) [m]")
plt.ylabel("y(t) [m]")
plt.axes().set_aspect('equal')
plt.savefig("earth_spiral_orbit.png", dpi=300)

fig3 = plt.figure()
plt.plot(arrays1[0],(arrays1[3]**2+arrays1[4]**2)**0.5,linewidth="0.25")
```

```
plt.xlabel("t [s]")
plt.ylabel("v [m/s]")
plt.savefig("earth_velocities.png", dpi=300)

import ephem

#Interplanetary Maneuvering

def Earth_pos(t):
    return R_E_orbit*np.cos((2*np.pi/T_E)*t),R_E_orbit*np.sin((2*np.pi/T_E)*t)

def Saturn_pos(t):
    return R_Sat_orbit*np.cos((2*np.pi/T_Sat)*t+theta_i_Sat),
    R_Sat_orbit*np.sin((2*np.pi/T_Sat)*t+theta_i_Sat)

def Get_angle(x,y):
    return np.atan(y/x)

#Euler-Richardson Scheme to obtain trajectory of spacecraft
def ER_SC(initial,t,t_stop):
    #initial parameters
    x,y,vx,vy,ax,ay=initial

    #arrays of each variable
    x_array = np.array([])
    y_array = np.array([])
    vx_array = np.array([])
    vy_array = np.array([])
    ax_array = np.array([])
    ay_array = np.array([])
    t_array = np.array([])

    i=0

    while (x**2+y**2)**0.5<=R_Sat_orbit:
        #compute acceleration
        ax = -G*(m_Sun*x)/(x**2+y**2)**1.5
        -G*(m_E*(x-Earth_pos(t*i)[0]))/((x-Earth_pos(t*i)[0])**2+
        (y-Earth_pos(t*i)[1])**2)**1.5
        -G*(m_Sat*(x-Saturn_pos(t*i)[0]))/((x-Saturn_pos(t*i)[0])**2+
        (y-Saturn_pos(t*i)[1])**2)**1.5

        ay = -G*(m_Sun*y)/(x**2+y**2)**1.5
        -G*(m_E*(y-Earth_pos(t*i)[1]))/((x-Earth_pos(t*i)[0])**2+
        (y-Earth_pos(t*i)[1])**2)**1.5
```

```
        -G*(m_Sat*(y-Saturn_pos(t*i)[1]))/((x-Saturn_pos(t*i)[0])**2+
        (y-Saturn_pos(t*i)[1])**2)**1.5

        #calculate middle terms
        x_mid = x+t/2*vx
        y_mid = y+t/2*vy
        vx_mid = vx+t/2*ax
        vy_mid = vy+t/2*ay
        ax_mid = -G*(m_Sun*x_mid)/(x_mid**2+y_mid**2)**1.5
        -G*(m_E*(x_mid-Earth_pos(t*i)[0]))/((x_mid-Earth_pos(t*i)[0])**2+
        (y_mid-Earth_pos(t*i)[1])**2)**1.5
        -G*(m_Sat*(x_mid-Saturn_pos(t*i)[0]))/((x_mid-Saturn_pos(t*i)[0])**2+
        (y_mid-Saturn_pos(t*i)[1])**2)**1.5
        ay_mid = -G*(m_Sun*y_mid)/(x_mid**2+y_mid**2)**1.5
        -G*(m_E*(y_mid-Earth_pos(t*i)[1]))/((x_mid-Earth_pos(t*i)[0])**2+
        (y_mid-Earth_pos(t*i)[1])**2)**1.5
        -G*(m_Sat*(y_mid-Saturn_pos(t*i)[1]))/((x_mid-Saturn_pos(t*i)[0])**2+
        (y_mid-Saturn_pos(t*i)[1])**2)**1.5


        #append to arrays
        t_array = np.append(t_array,t*i)
        x_array = np.append(x_array,x)
        y_array = np.append(y_array,y)
        vx_array = np.append(vx_array,vx)
        vy_array = np.append(vy_array,vy)
        ax_array = np.append(ax_array,ax)
        ay_array = np.append(ay_array,ay)

        #update vx, vy, x, and y
        x = x + t*vx_mid
        y = y + t*vy_mid
        vx = vx + t*ax_mid
        vy = vy + t*ay_mid

        i=i+1

    print("Time of Flight: ", t_array[-1])
    print("Final speed:",(vx_array[-1]**2+vy_array[-1]**2)**0.5)
    return(t_array, x_array, y_array, vx_array, vy_array, ax_array, ay_array)

launch_angle = np.radians(78)
launch_speed = 10.7*1000 #m/s
x0_sc = R_E_orbit+924648*1000#radius of earth's orbit+sphere of influence
y0_sc = 0
```

```
vx0_sc = launch_speed*np.cos(launch_angle) #m/s
vy0_sc = 30*1000+launch_speed*np.sin(launch_angle) #m/s
ax0_sc = 0
ay0_sc = 0
ini = [x0_sc,y0_sc,vx0_sc,vy0_sc,ax0_sc,ay0_sc]

arrays = ER_SC(ini,86400*0.25,t_meas*365*86400*2)
final_angle = math.atan(arrays[2][-1]/arrays[1][-1])
print("True initial position",arrays[1][-1],arrays[2][-1])

#Plotting
fig = plt.figure()
plt.plot(E_x,E_y,label = "Earth orbit")
plt.plot(Sat_x,Sat_y,label = "Saturn orbit")
plt.plot(arrays[1],arrays[2],label = "Spacecraft Position")
plt.plot(arrays[1][-1],arrays[2][-1],'bo', markersize=5)
plt.xlim(-1.1*R_Sat_orbit,1.1*R_Sat_orbit)
plt.ylim(-1.1*R_Sat_orbit,1.1*R_Sat_orbit)
plt.xlabel("x(t) [m]")
plt.ylabel("y(t) [m]")
plt.legend(loc ="lower right")
plt.axes().set_aspect('equal')
plt.savefig("interplanetary_trajectory.png", dpi=300)

fig2 = plt.figure()
plt.plot(arrays[0],(arrays[3]**2+arrays[4]**2)**0.5)
plt.xlabel("t [s]")
plt.ylabel("v [m/s]")
plt.savefig("interplanetary_velocity.png", dpi=300)
```

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